

Multimodality @ Hugging Face

Sergio Paniego Blanco (@sergiopaniego)

Machine Learning Engineer @ Hugging Face



Content

- 
- What's HF?
 - What's multimodality?
 - Why is it interesting now?
 - How do we choose the best?
 - How do we build/use with it?
- 

Hi, I'm Sergio 🖐️

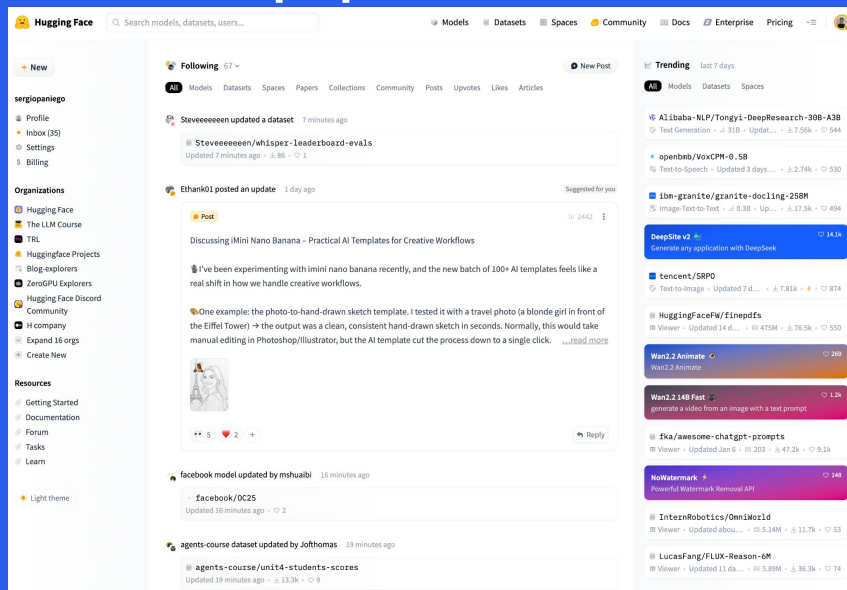
- Machine Learning Engineer @ Hugging Face
 - Multimodality, Post-Training & Agents
- PhD in AI
- Open source and AI → GSoC
- Teaching and Sharing



What's Hugging Face? 🤗

AI open platform (+community) where AI developers share (HF Hub):

- Models, datasets, apps...
- Posts, blogs, courses, papers, leaderboards...



2.1M

public models

500K+

public datasets

1M+

daily downloads

1M+

daily visitors

What's Hugging Face? 🤗

Set of open source projects to empower AI development and advancement



Transformers



Datasets



Trackio



Hub



30+

Libraries



Transformers.js



Diffusers

timm



Accelerate



PEFT



smolagents

Open vs Closed



	Open	Closed / Proprietary
Security	Models can be self-hosted , data stays in your environment	Models cannot be self-hosted. Data is sent outside your environment to vendor
Control	The timing and nature of updates are controlled by you	Updates and changes to performance can happen without notice
Customization	Full source code access to customize the model for your needs	Limited ability to customize for your needs
Transparency	Inspect code and data provides better auditability and understandability	No ability to audit or understand performance
Cost	Typical lower long term cost due to smaller model size	larger model size and proprietary premium often balanced by decreased cost from server-side optimization
Latency	Lower latency due to on premise and smaller model sizes	Often greater latency due to larger model sizes + API latency
Quality	No single approach is best. Each use case will vary. Proprietary is typically closer to the frontier of performance .	
Examples	 OpenAI  Meta  deepseek  MISTRAL AI  Microsoft	 OpenAI ANTHROPIC

How HF democratizes ML/AI



ML engineers who
train models from
scratch

Software engineers
who use inference
APIs, ML engineers
who fine-tune
models

Anyone who
can use a GUI



Datasets



Models



Spaces



Used everywhere in the AI world

15,000+ startups and enterprises



Open source contributors



**Hugging
Face**

Cloud partners



Hardware partners



On-prem partners



What's multimodality?

Previously:

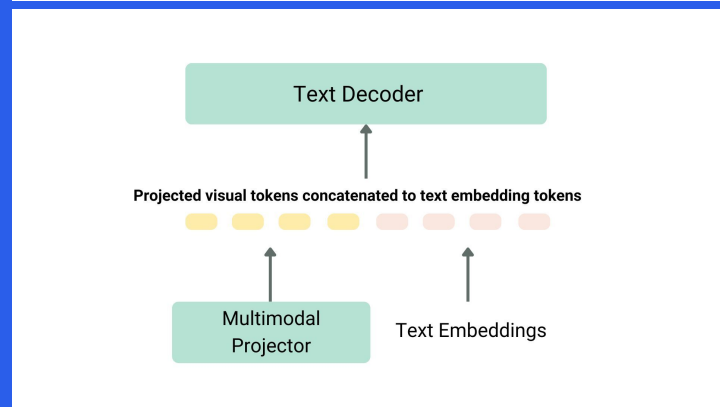
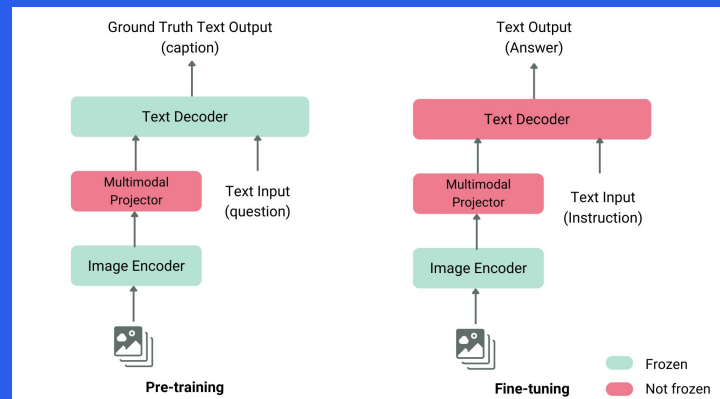
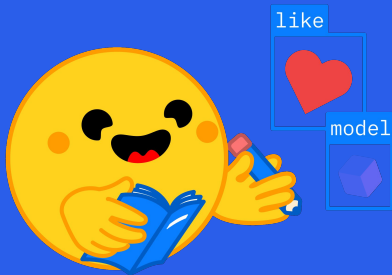
- Text-to-text  → 

Combining 2 or more modalities:

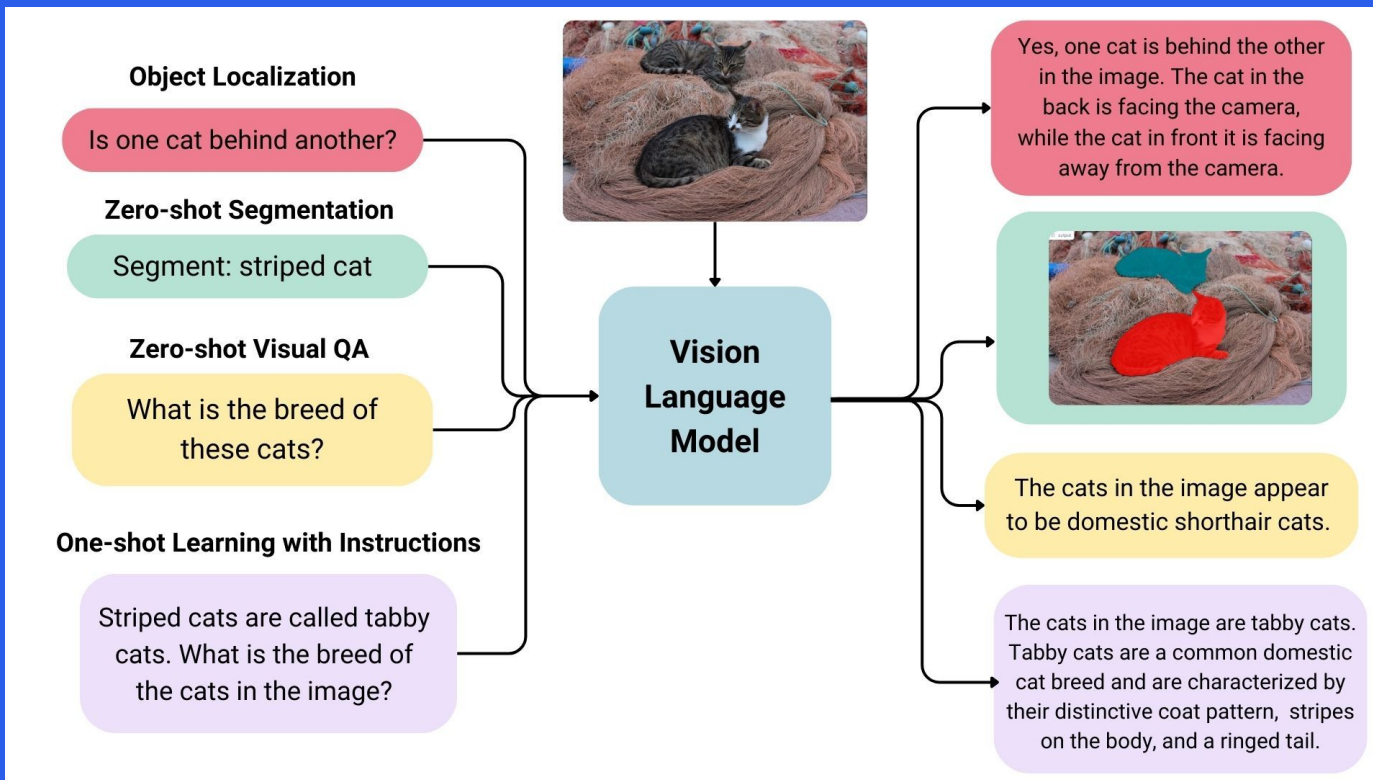
- Text 
- Image 
- Audio 
- Video 

Vision Language Models
(images+text)

Omni Models (any combination)



Vision Language Models



Multimodal tasks in Hugging Face

Multimodal



Any-to-Any

7,736 models

Contribute 



Audio-Text-to-Text

126 models



**Document
Question
Answering**

233 models



**Visual
Document
Retrieval**

89 models



Image-Text-to-Text

6,583 models



Video-Text-to-Text

178 models



**Visual Question
Answering**

516 models

hf.co/tasks

Image-text-to-text (VLMs)

Inputs



Text Prompt

Describe the position of the bee in detail.

Image-Text-to-Text Model

Output

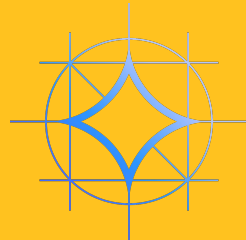
Answer

The bee is sitting on a pink flower, surrounded by other flowers. The bee is positioned in the center of the flower, with its head and front legs sticking out.

<https://huggingface.co/tasks/image-text-to-text>

Image-text-to-text (VLMs)

- Qwen3-VL/Qwen2.5-VL
- SmolVLM2
- Gemma 3n
- Pixtral
- GLM-4.5V
- Moondream 3
- Llama 4
- InternVL3.5
- Kimi-VL-A3B
- MiniCPM-V-4_5
- DeepSeek-VL2

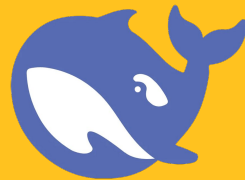
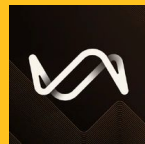
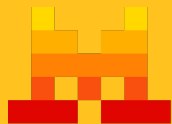


GLM-V



InternVL

SmolVLM2



HF collection

Vision Language Models: 2025 Update

updated May 12

This collection includes all the models, datasets and Spaces mentioned in the blog Vision Language Models: 2025 Update 

Qwen/Qwen2.5-Omni-7B

Any-to-Any · 11B · Updated Apr 30 · 176k · 1.79k

Qwen2.5 Omni 7B Demo 🏆

349

Generate text and speech from audio, video, and text inputs

Qwen2.5-Omni Technical Report

Paper · 2503.20215 · Published Mar 26 · 165

openbmb/MiniCPM-o-2_6

Any-to-Any · 9B · Updated Jun 20 · 155k · 1.25k

deepseek-ai/Janus-Pro-7B

Any-to-Any · Updated Feb 1 · 190k · 3.5k

Chat With Janus-Pro-7B 🌟

2k

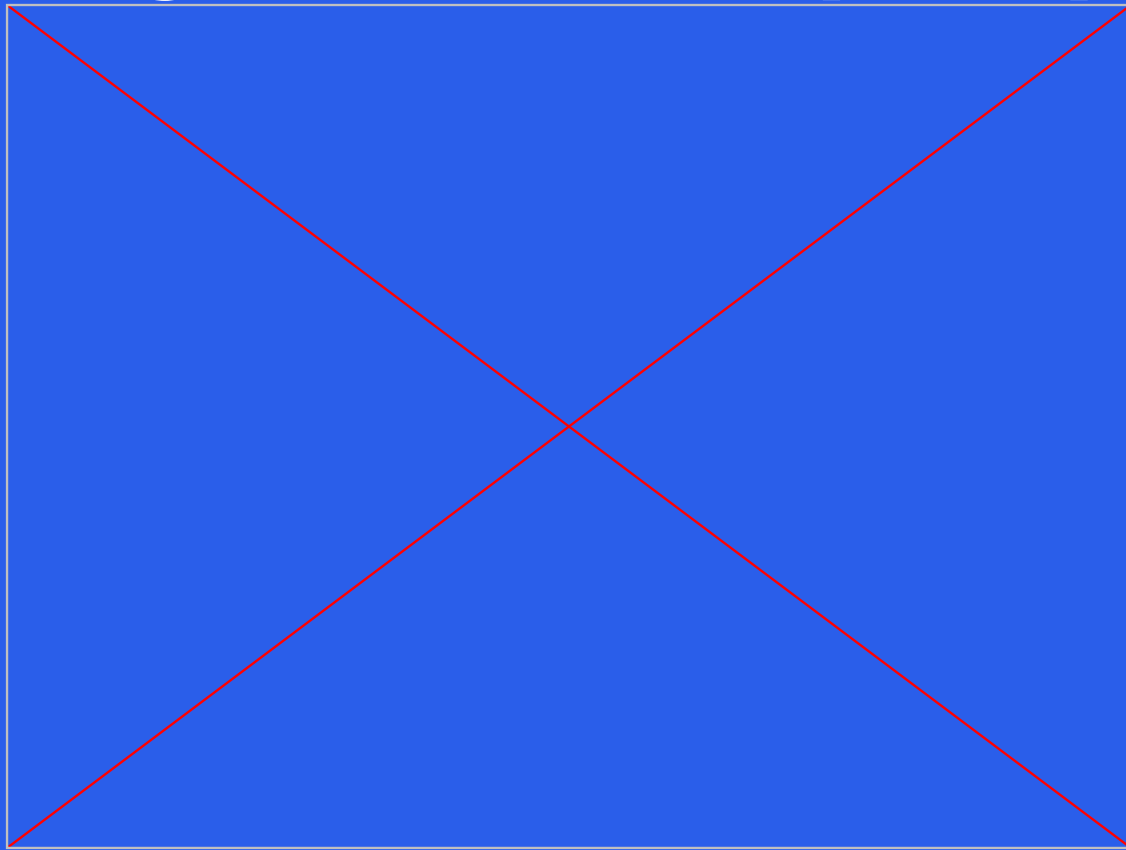
A unified multimodal understanding and generation model.

Janus-Pro: Unified Multimodal Understanding and Generation with Data and Model Scaling

Paper · 2501.17811 · Published Jan 29 · 6



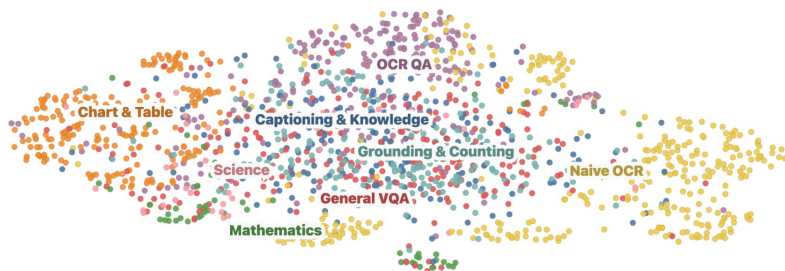
Image-text-to-text (VLMs)



https://huggingface.co/spaces/sergiopaniego/vlm_object_understanding

Not only VLM models!

FineVision: Open Data Is All You Need



A new open dataset for data-centric training of Vision Language Models

AUTHORS

[Luis Wiedmann](#)
[Orr Zohar](#)
[Andi Marafioti](#)
[Amir Mahla](#)
[Thibaud Frere](#)
[Leandro von Werra](#)

AFFILIATION

[Hugging Face](#)

PUBLISHED

Sep 4, 2025

PDF

[Download PDF](#)

Table of Contents

Why this dataset?

How did we build FineVision?

Experimental Setup

Experiments

Conclusion

Today, we release **FineVision**, a new multimodal dataset with **24 million samples**. We created FineVision by collecting over **200 datasets** containing **17M images**, **89M question-answer turns**, and **10B answer tokens**, totaling **5TB of high-quality data**. Additionally, we extensively processed all datasets to unify their format, clean them of duplicates and poor data, and rated all turns using 32B VLMs across 4 qualitative metrics with a score from 1-5 to enable the construction and study of individual training mixtures.

huggingface.co/datasets/HuggingFaceM4/FineVision

<https://huggingface.co/spaces/HuggingFaceM4/FineVision>

Document/Visual QA

Inputs

Question

What is the idea behind the consumer relations efficiency team?

July 31, 2003

Consumer Relations Efficiency Team

<u>Team:</u>	Nancy Rowland	Sherrod McManamy	Veronica Watson
	Gary Hicks <th>Sue Montgomery</th> <th>Yvonne Willard</th>	Sue Montgomery	Yvonne Willard
	John Haddock <td>Dee O'Burne <td></td> </td>	Dee O'Burne <td></td>	
	Crab Hunter <td>Doreen Wallop <td></td> </td>	Doreen Wallop <td></td>	

Objective: Balance cost efficiency with quality customer service

Our Process:

- Examined *Consumer Relations cost structure:*
 - Woe Interactive - Advertiser
 - Bellcity - Live Rep
 - VA - Live Rep
 - Glostream
 - ATAT
 - Broad Changelinks
 - Vantis

<u>Reasons for Consumer Contact</u>	%
Transfer Customer Order Line	2%
Order Status	3%
Marketing In Request (add, change, update, add, target change)	24%
Product/Service Question (add, change, update, questions, sales, Web, online, etc)	25%
Transfer Issues	4%
Low account ID, conversion, full-time	4%
2.P.P Issues - card, reporter, statement issues	1%

Output

Answer

Balance cost efficiency with quality customer service

Inputs



Question

What is in this image?

**Visual
Question
Answering
Model**

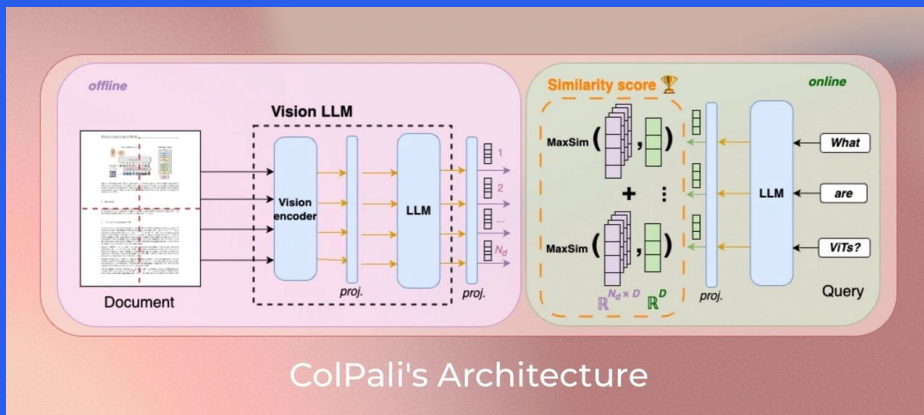
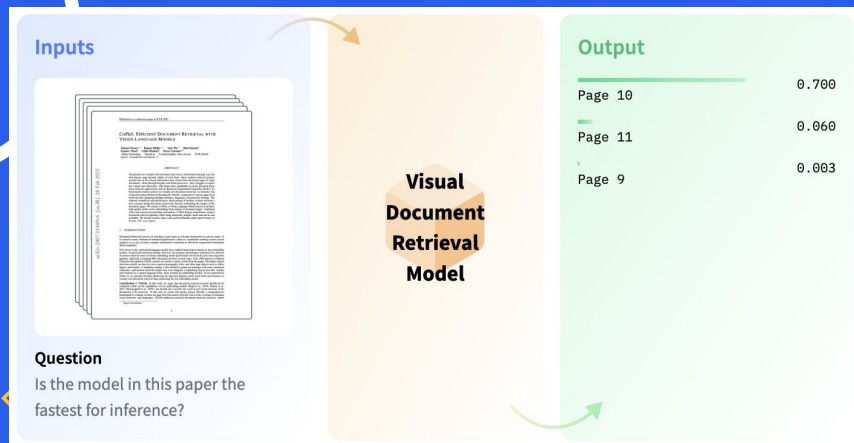
Output

elephant	0.970
elephants	0.060
animal	0.003

<https://huggingface.co/tasks/document-question-answering>
<https://huggingface.co/tasks/visual-question-answering>

Visual Document Retrieval (multimodal RAG)

ColPali, ColQwen2...



<https://huggingface.co/tasks/visual-document-retrieval>

Any-to-Any (Omni)



Inputs



Text Prompt

What is the significance of this place?

Any-to-Any Model

Output

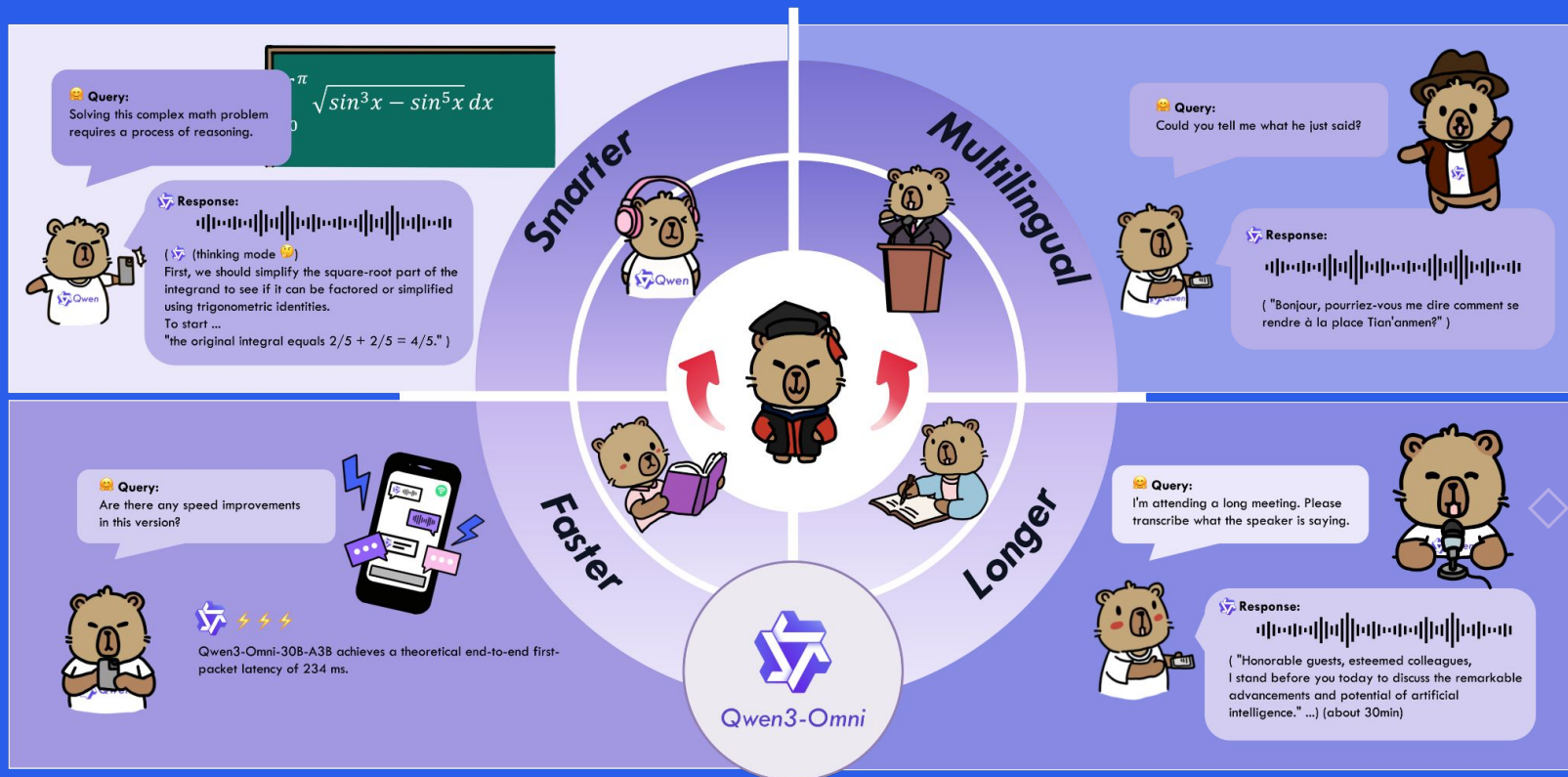
Generated Text

The place in the picture is Osaka Castle, located in Osaka, Japan. Osaka Castle is a historic castle that was originally built in the 16th century by Toyotomi Hideyoshi, a powerful warlord of the time. It is one of the most famous landmarks in Osaka and is known for its distinctive white walls and black roof tiles. The castle has been rebuilt several times over the centuries and is now a popular tourist attraction, offering visitors a glimpse into Japan's rich history and culture.

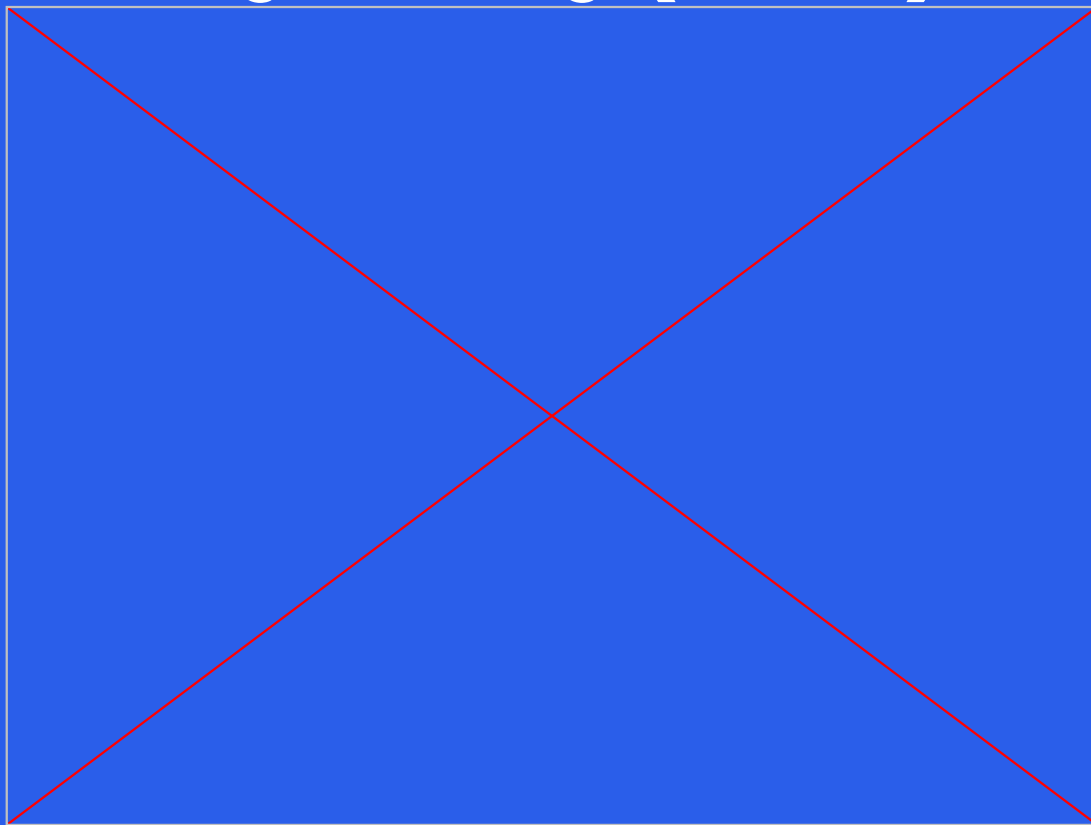
▶ 0:00 / 0:29 — 🔊 ⋮

<https://huggingface.co/tasks/any-to-any>

Any-to-Any (Omni)



Any-to-Any (Omni)

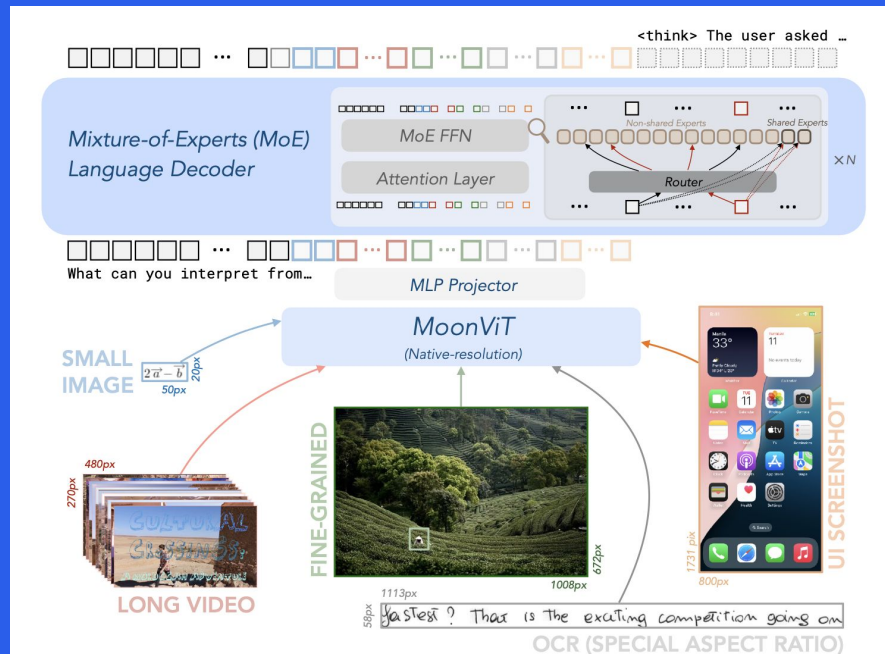


<https://huggingface.co/spaces/Qwen/Qwen3-Omni-Demo>



Trends

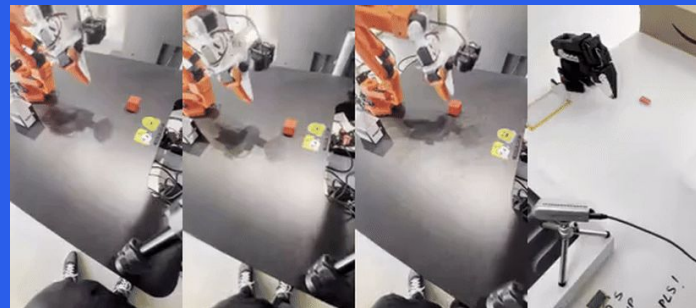
- Reasoning models:
 - QVQ-72B-preview
 - Kimi-VL-A3B-Thinking
- Smol models
 - Gemma 3n
 - smoVLM2
 - Qwen2.5-VL-3B-Instruct
 - Moondream 3B
 - PerceptronAI/Issac-0.1
- MoEs





Trends

- **Vision-Language-Action (VLAs)**
 - SmoIVLA, GROOT N1, ShowUI, SimLingo
- **Specialized models** (Object Detection, Segmentation, Counting)
 - PaliGemma, Molmo, moondream3, Issac0.1



point to candies in the image

Molmo
→



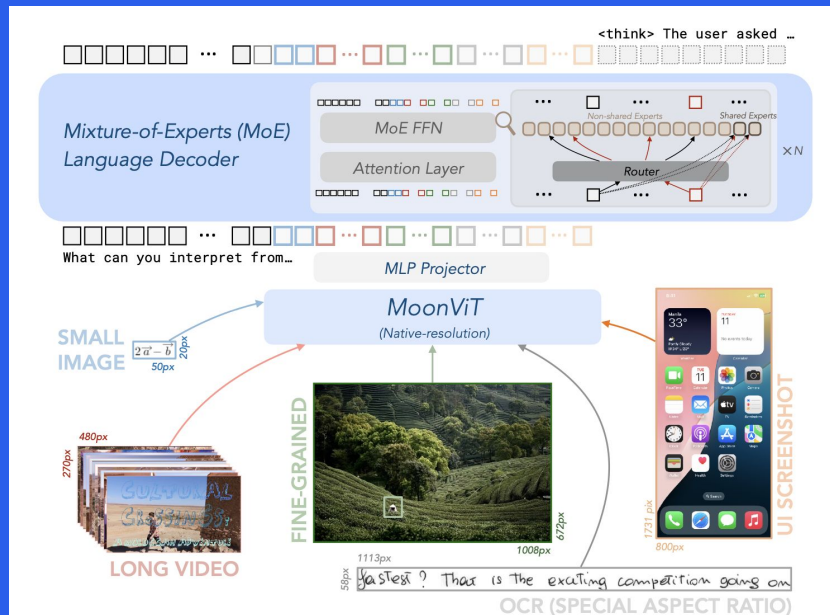
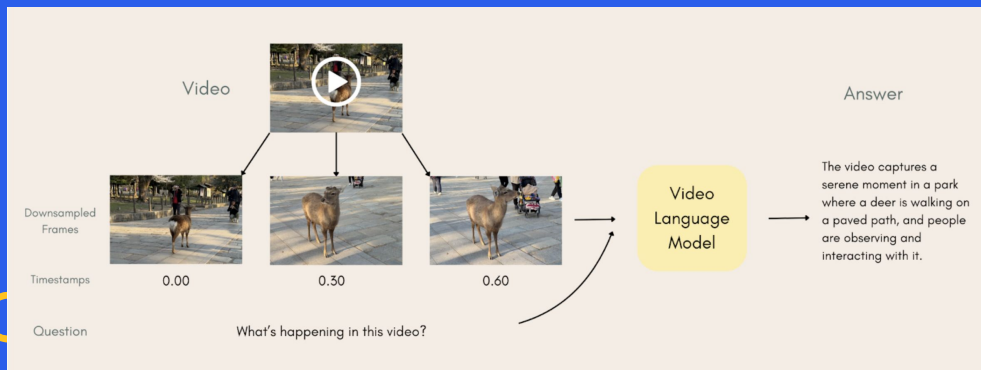
• the candies in the image

Counting the the candies in the image shows a total of 5.



Trends

- **Agentic tasks:**
 - Kimi-VL-A3B-Thinking
 - Qwen2.5-VL-32B
- **Video Language Models:**
 - SmoIVLM2, LongVU, Qwen2.5VL, Gemma3n



Choosing the best

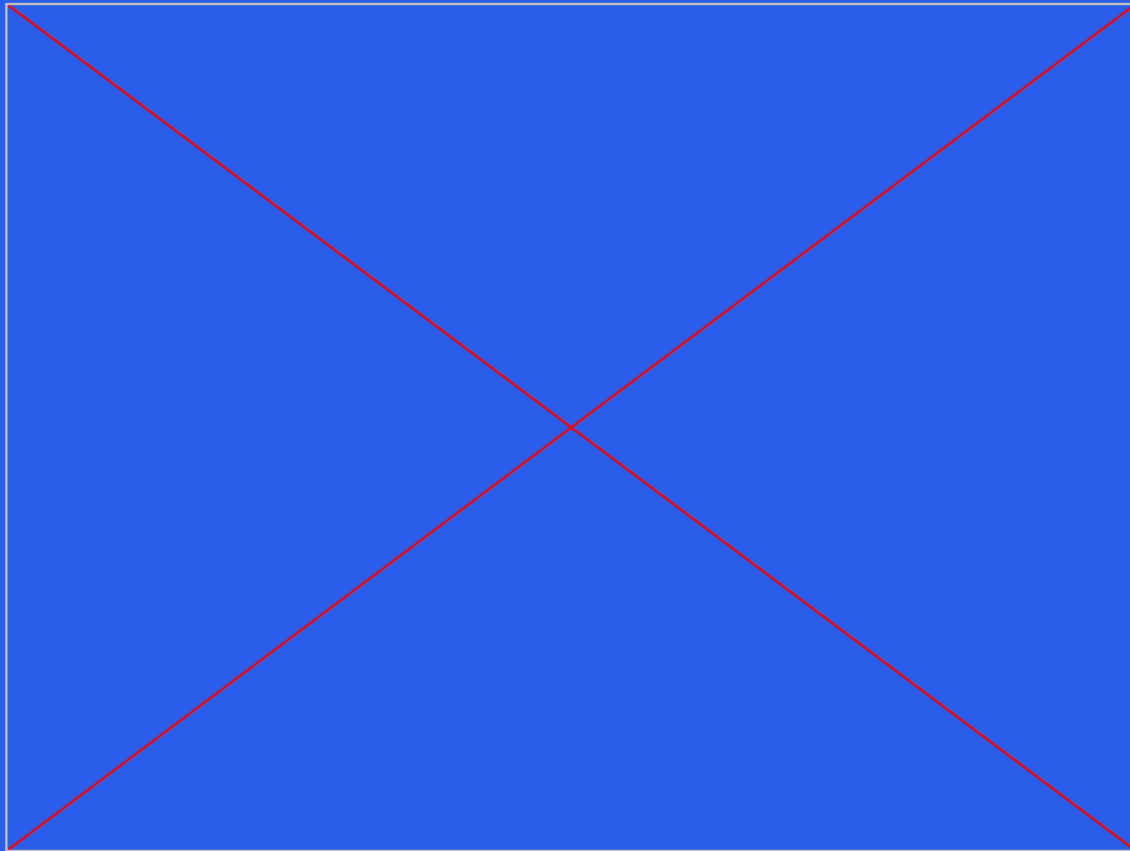


- **General:** MMMU-Pro, MMT-Bench...
- **Specialized benchmarks**
- **OpenVLM Leaderboard**
- **Trade-offs:** size 📏, speed ⚡, quality ★, licensing 🔒

Model Name		Model Size						Model Type							
Input the Model Name (fuzzy, case insensitive)		☐ <4B	☐ 4B-10B	☐ 10B-20B	☐ 20B-40B	☒ >40B	☐ Unknown	☒ API	☒ OpenSource						
Rank	Method	Param (B)	Language Model	Vision Model	Eval Date	Avg Score	Avg Rank	MMBench_V11	MMStar	MMMU_VAL	MathVista	OCRBench	AI2D	HallusionBench	MMVet
1	InternVL3-78B	78.4	Qwen2.5-72B	InternViT-6B-v2.5	2025/04/14	79.1	5.88	87.7	73.4	72.2	79	908	89.8	59.1	80.7
2	InternVL2.5-78B-MPO	78	Qwen-2.5-72B	InternViT-6B-v2.5	2024/12/29	77	12.88	87.7	72.1	68.2	76.6	909	89.2	58.1	73.5
3	Qwen2.5-VL-72B	73.4	Qwen2.5-72B	QwenViT	2025/02/02	76.1	19	87.8	70.5	68.2	74.2	882	88.5	54.6	76.9
4	BailingMM-Pro-0120	81	Bailing-80B	SigLIP-400M	2025/01/25	75.2	21.62	87.1	69.4	59.2	72.3	894	87.2	59.4	77.6
5	Qwen2-VL-72B	73.4	Qwen2-72B	QwenViT	2024/10/28	74.8	24.38	85.9	68.6	64.3	69.7	888	88.3	58.7	73.9
6	Qwen-VL-Max-0809	72	Qwen2-72B	QwenViT	2024/09/12	74.4	26.5	85.8	69.2	64.6	68.3	881	88.1	59.2	72.3
7	InternVL2-Llama3-76B	76	Llama-3-70B-Instruct	InternViT-6B	2024/07/19	71	46.25	85.5	67.1	58.3	65.6	842	87.6	55.4	64.4
8	InternVL2-40B	40	Nous-Hermes-2-Yi-34B	InternViT-6B	2024/07/11	69.7	54	85	64.7	55.2	64.3	833	86.8	56.5	61.8
9	InternVL2.5-78B	78	Qwen-2.5-72B	InternViT-6B-v2.5	2024/12/10	68.3	51.75	87.5	69.5	70	70.6	853	89.1	57.4	16.9

https://huggingface.co/spaces/opencompass/open_vlm_leaderboard

Choosing the best (Vibes)



<https://huggingface.co/spaces/visionLMsftw/comparevlms>

Everything done by a single

How Well Does GPT-4o Understand Vision? Evaluating Multimodal Foundation Models on Standard Computer Vision Tasks

Rahul Ramachandran, Ali Garjani, Roman Bachmann,

Andrei Atanov*, Oğuzhan Fatih Kar*, Amir Zamir*

Swiss Federal Institute of Technology (EPFL)

* Equal technical advising.

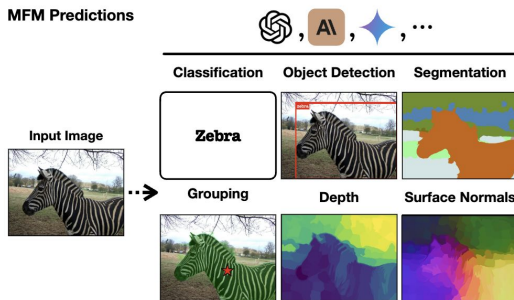
✕ Paper

🔗 Code

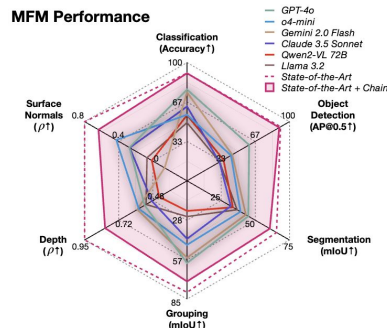
📄 Slides ▾

🐦 Twitter

MFM Predictions



MFM Performance



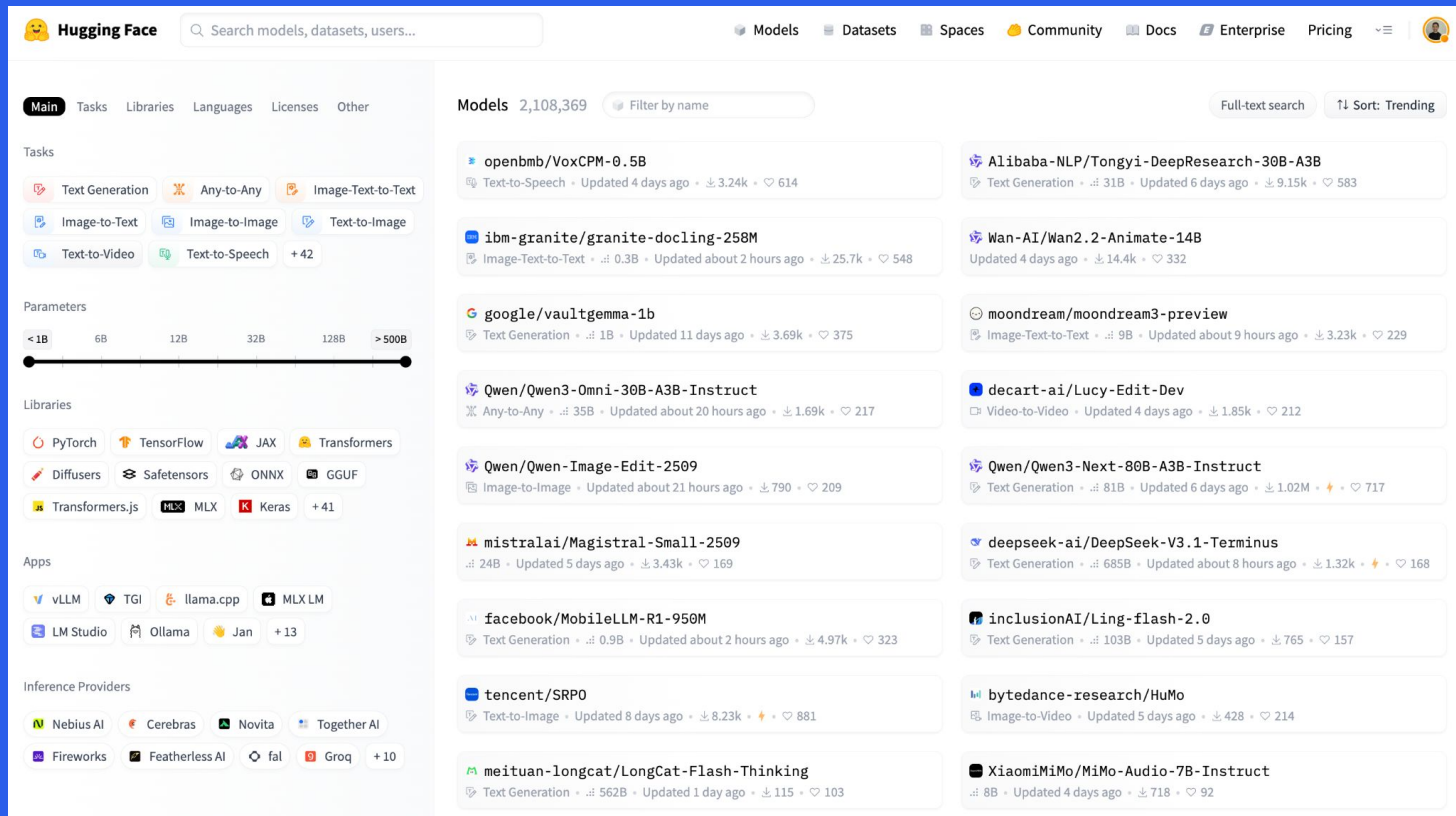
<https://huggingface.co/papers/2507.01955>

How do we build with it?

Find model → Customize → Deploy



Find the model



The screenshot shows the Hugging Face website's 'Models' section. The left sidebar contains navigation links for 'Main', 'Tasks', 'Libraries', 'Languages', 'Licenses', and 'Other'. Under 'Tasks', there are buttons for 'Text Generation', 'Any-to-Any', 'Image-Text-to-Text', 'Image-to-Text', 'Image-to-Image', 'Text-to-Image', 'Text-to-Video', 'Text-to-Speech', and '+42'. The 'Parameters' section features a slider from '<1B' to '>500B'. The 'Libraries' section lists various frameworks like PyTorch, TensorFlow, JAX, Transformers, etc. The 'Apps' section shows vLLM, TGI, llama.cpp, etc. The 'Inference Providers' section lists Nebius AI, Cerebras, etc. The main content area displays a list of models with filters and sorting options. The models listed include:

- openbmb/VoxCPM-0.5B**: Text-to-Speech • Updated 4 days ago • 3.24k • 614
- Alibaba-NLP/Tongyi-DeepResearch-30B-A3B**: Text Generation • 31B • Updated 6 days ago • 9.15k • 583
- ibm-granite/granite-docling-258M**: Image-Text-to-Text • 0.3B • Updated about 2 hours ago • 25.7k • 548
- Wan-AI/Wan2.2-Animate-14B**: Updated 4 days ago • 14.4k • 332
- google/vaultgemma-1b**: Text Generation • 1B • Updated 11 days ago • 3.69k • 375
- moondream/moondream3-preview**: Image-Text-to-Text • 9B • Updated about 9 hours ago • 3.23k • 229
- Qwen/Qwen3-Omni-30B-A3B-Instruct**: Any-to-Any • 35B • Updated about 20 hours ago • 1.69k • 217
- decart-ai/Lucy-Edit-Dev**: Video-to-Video • Updated 4 days ago • 1.85k • 212
- Qwen/Qwen-Image-Edit-2509**: Image-to-Image • Updated about 21 hours ago • 790 • 209
- Qwen/Qwen3-Next-80B-A3B-Instruct**: Text Generation • 81B • Updated 6 days ago • 1.02M • 717
- mistralai/Magistral-Small-2509**: 24B • Updated 5 days ago • 3.43k • 169
- deepseek-ai/DeepSeek-V3.1-Terminus**: Text Generation • 685B • Updated about 8 hours ago • 1.32k • 168
- facebook/MobileLLM-R1-950M**: Text Generation • 0.9B • Updated about 2 hours ago • 4.97k • 323
- inclusionAI/Ling-flash-2.0**: Text Generation • 103B • Updated 5 days ago • 765 • 157
- tencent/SRPO**: Text-to-Image • Updated 8 days ago • 8.23k • 881
- bytedance-research/HuMo**: Image-to-Video • Updated 5 days ago • 428 • 214
- meituan-longcat/LongCat-Flash-Thinking**: Text Generation • 562B • Updated 1 day ago • 115 • 103
- XiaomiMiMo/MiMo-Audio-7B-Instruct**: 8B • Updated 4 days ago • 718 • 92

<https://huggingface.co/models>

Use it with 🧡 Transformers

The Transformers Library: standardizing model definitions

Published May 15, 2025

[Update on GitHub](#)



Lysandre
lysandre

Follow



Arthur Zucker
ArthurZ

Following



Pedro Cuenca
pcuenca

Following



Julien Chaumond
julien-c

Following

TLDR: Going forward, we're aiming for Transformers to be the pivot across frameworks: if a model architecture is supported by transformers, you can expect it to be supported in the rest of the ecosystem.

Transformers was created in 2019, shortly following the release of the BERT Transformer model. Since then, we've continuously aimed to add state-of-the-art architectures, initially focused on NLP, then growing to Audio and computer vision. Today, transformers is the default library for LLMs and VLMs in the Python ecosystem.



<https://huggingface.co/blog/transformers-model-definition>



Use it with 🙌 Transformers

```
from transformers import pipeline

pipe = pipeline("image-text-to-text")
messages = [
    {
        "role": "user",
        "content": [
            {
                "type": "image",
                "image": "https://huggingface.co/datasets/huggingface/documentation-images/resolve/main/bee.jpg",
            },
            {
                "type": "text",
                "text": "Describe this image.",
            },
        ],
    }
]

outputs = pipe(text=messages, max_new_tokens=60, return_full_text=False)
print(outputs[0]["generated_text"])
# The image depicts a close-up view of a vibrant and colorful flower. The flower in focus is a large, pink, five-petal
# flower with a prominent yellow center, which is characteristic of many species of the genus Dianthus. The petals are
# broad and slightly curved, giving the flower a full
```

<https://huggingface.co/docs/transformers/index>



Use it with 🧑🏻💻 Transformers

```
#pip install qwen-vl-utils[decord]==0.0.8
from transformers import AutoProcessor, AutoModelForImageTextToText
from qwen_vl_utils import process_vision_info
import torch

model_name = "Qwen/Qwen2.5-VL-3B-Instruct"
processor = AutoProcessor.from_pretrained(model_name)
model = AutoModelForImageTextToText.from_pretrained(model_name, torch_dtype="auto", device_map="auto")

messages = [
    {
        "role": "user",
        "content": [
            {"type": "image", "image": "https://huggingface.co/datasets/huggingface/documentation-images/resolve/main/bee.jpg"},
            {"type": "text", "text": "Describe this image."},
        ],
    },
]

text = processor.apply_chat_template(
    messages, tokenize=False, add_generation_prompt=True
)
image_inputs, _ = process_vision_info(messages)
inputs = processor(
    text=[text],
    images=image_inputs,
    padding=True,
    return_tensors="pt",
)
inputs = inputs.to("cuda")

generated_ids = model.generate(**inputs, max_new_tokens=128)
generated_ids_trimmed = [out_ids[len(in_ids):] for in_ids, out_ids in zip(inputs.input_ids, generated_ids)]
output_text = processor.batch_decode(generated_ids_trimmed, skip_special_tokens=True, clean_up_tokenization_spaces=False)
print(output_text[0])

# The image depicts a close-up view of several flowers in a garden setting. The flowers are predominantly pink with some having yellow centers, and they appear to be cosmos or similar types of daisy-like plants. The petals are delicate and slightly ruffled, giving them a soft appearance. Some of the flowers have fully bloomed, while others are still in bud form.
```



Customize with **TRL**

TRL allows you to fine-tune VLMs using different methods:
SFT, DPO, MPO, GRPO, GSPO, RLOO, and Online DPO.
PEFT and quantization support

```
from trl import SFTConfig, SFTTrainer
from datasets import load_dataset

trainer = SFTTrainer(
    model="Qwen/Qwen2.5-VL-3B-Instruct",
    args=SFTConfig(max_length=None),
    train_dataset=load_dataset("trl-lib/llava-instruct-mix", split="train"),
)
trainer.train()
```

Deploy model

Inference and serving frameworks

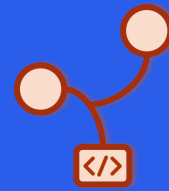
Inference Providers (TGI)

Hugging Face Endpoints 



 VLLM

LLaMA 



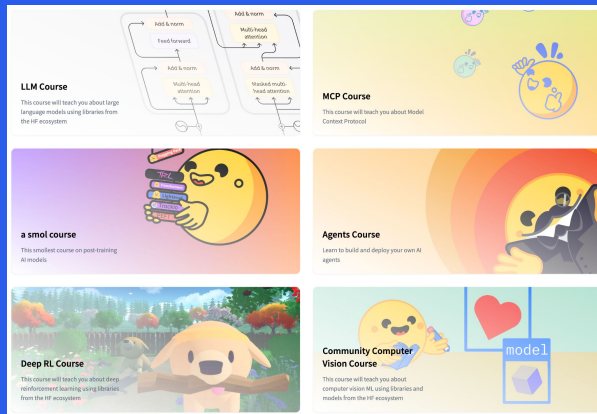
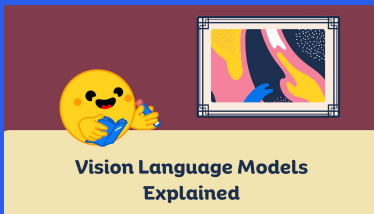
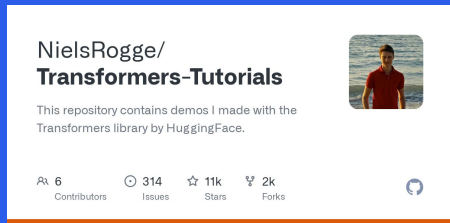
SG L



Resources



- hf.co/blog
- hf.co/learn
- [Transformers-Tutorials repo](https://github.com/NielsRogge/Transformers-Tutorials)
- [smol-vision repo](https://github.com/NielsRogge/smol-vision)



Closing



- Hugging Face: hub, libraries, and community for AI
- Multimodal AI: combining text, images, audio, & video
- Benchmarks & evaluation
- Tools for use, customization, and deployment: Transformers, TRL, Spaces...

Thanks!

Sergio Paniego Blanco (@sergiopaniego)

Machine Learning Engineer @ Hugging Face

